ch 6) Replacement Analysis

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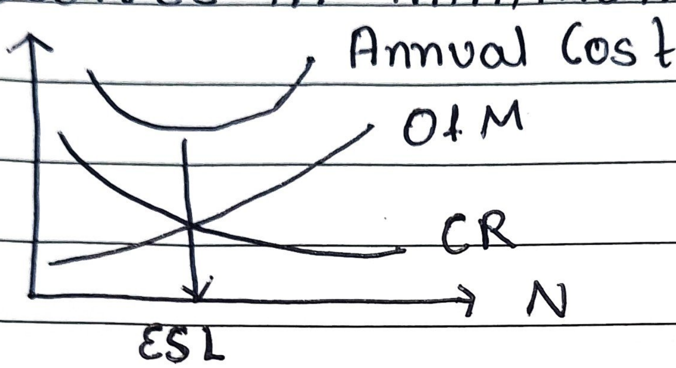
6.1) Replacement Strategy

6.2) Economic Service of life (ESL)

- Remaining useful life that resolves in minimum AEC



- $AEC = CR + O\&M$ (Min → ESL)

- Capital Recovery (CR)

$$CR = I(A/P, i\%, n) - S(A/F, i\%, n)$$

- Operating Cost (O&M)

$$O\&M = \sum^{N} OC\,(P/F, i\%, n)\,(A/P, i\%, N)$$

$$O\&M = OC\left(\frac{1 - \left(\frac{1+g}{1+i}\right)^n}{i - g}\right)$$

6.3) finite Planning Horizon

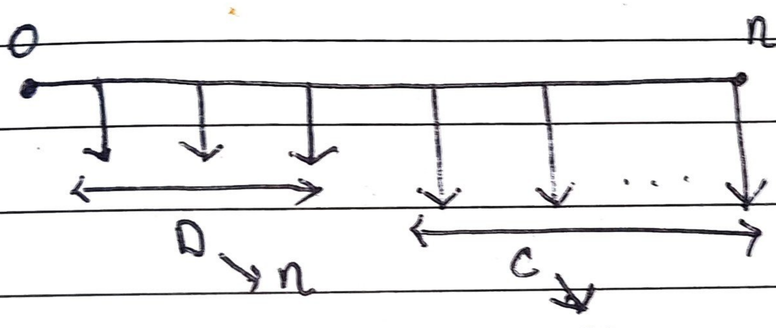
• Short cut Method

Given → AEC (C/D), MARR-$i\%$, Planning Period

ESL of D Combine $(D, \_)\ \underbrace{(C, \_)}$

ESL of C sum = plan period

Make different options



$$PW = D(P/A, i\%, \underline{n}) + C(P/A, i\%, m)(P/F, i\%, \underline{n})$$

[PYQs]

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1. Defender
→ It is assest being replaced

2. Challenger
→ It is new asset that is purposed to replace defender (old one)

3. Replacement Analysis
→ Process of deciding to choose b/w challenger & defender

4. AEC (Annual Equivalent Cost)
→ AEC = CR + O&M

5. Sunk Cost
→ Cost that is spent & can't be recovered

6. Need of Replacement
→ Deterioration
→ Absolensce
→ Depletion
→ Impairments